

Review

Review of Selected PCMs and Their Applications in the Industry and Energy Sector

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Abstract: The growing global energy demand, coupled with environmental concerns, necessitates the development of efficient energy storage technologies. Phase Change Materials (PCMs) have emerged as a promising solution for thermal energy storage (TES) due to their ability to store and release latent heat during phase transitions. This review provides a comprehensive analysis of PCMs, exploring their classifications based on phase transition types, chemical compositions, and thermophysical properties. Additionally, the review highlights advancements in developing organic, inorganic, and metallic PCMs and evaluates their potential applications in sectors such as solar energy, construction, and automotive industries. Methodologies include a detailed examination of the strengths, limitations, and solutions to challenges such as low thermal conductivity, phase separation, and supercooling. The results summarize the diverse applications of PCMs, emphasizing their critical role in enhancing energy efficiency and sustainability. The review concludes with recommendations for overcoming current limitations and future directions for PCM research and technology integration across various industries.

Keywords: phase change materials (PCMs); thermal energy storage; applications



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1. Introduction

According to data from the International Energy Agency (IEA), between 1990 and 2017, there was an increase of about 55% in final energy consumption worldwide [1]. It is also worth noting that the structure of energy consumption is dominated by fossil fuels, the processing of which to obtain useful forms of energy, e.g., electricity is associated with the generation of heat as well as various substances, e.g., greenhouse gases, which can have an adverse impact on the environment. On the other hand, an increase in the share of renewable energy sources (RES) can be observed, but their main disadvantage is their unpredictability and the fact that the energy demand and the production capacity of RES often do not overlap in time. As a result, it seems necessary to use energy storage, including thermal energy storage (TES), through which surplus energy can be accumulated and used during increased demand.

Heat storage can take place in three ways, depending on the phenomenon used. The first method uses the specific sensible heat of the storage material and involves raising its temperature. Materials that are used to store sensible heat include, e.g., water or concrete [2]. The second method of heat storage is based on the use of reversible phase transitions during which so-called latent heat is released or absorbed. Materials that undergo such transitions

are called phase-change materials (PCMs). A third way of storing heat is called thermo-chemical energy storage and involves the use of reversible endo- and exothermic chemical reactions [2].

2. Classification of PCMs

As already mentioned, PCMs are materials that undergo reversible phase transitions during which heat is released or absorbed. PCMs are usually classified based on the type of phase transition that takes place or on their chemical composition [3], but they can also be classified based on their melting temperature range and associated application possibilities [4].

Regardless of the type, PCMs should meet a number of criteria, which include [3,5,6]:

- A melting point suitable for the application;
- The high heat of phase transition and specific heat;
- High density;
- High thermal conductivity coefficient;
- No change in chemical composition during the phase transition (congruent melting);
- No change in volume during phase transition;
- No supercooling;
- High chemical stability;
- No change in material properties after multiple phase transitions;
- No chemical reaction with the container material;
- Non-toxic;
- Non-flammable;
- Low cost;
- Wide-scale availability.

2.1. Classification by the Type of Phase Transition

The first classification criterion, i.e., by the type of phase transition, distinguishes materials that undergo the following transitions: solid-solid, solid-liquid, liquid-gas and solid-gas [3]. During solid-solid transitions, heat release or absorption occurs due to a change in the crystalline form of the material [6]. An undisputed advantage of these types of materials is that there is little or no change in volume during the transition, but the main disadvantage is the relatively low latent heat [6]. In contrast, solid-gas and liquid-gas transitions are disadvantageous from a practical point of view because the resulting gas occupies a large volume [6]. Therefore, most attention is devoted to PCMs that undergo solid-liquid transitions, and this study also focuses on this group of materials.

2.2. Classification by Chemical Composition

According to the second classification criterion, there are two main groups of PCMs that undergo solid-liquid transitions: organic and inorganic [3,5,7]. Organic materials mainly include paraffins and fatty acids [3,7], while inorganic materials include salts, salt hydrates and metals or their alloys [4,8,9], as shown schematically in Figure 1. It should be noted that the division adopted only includes substances that are most commonly considered in the literature as PCMs, but other substances, such as, e.g., alcohols [10,11] or fatty acid esters [4], can also act as PCMs. Furthermore, the abovementioned materials can be used to form mixtures or eutectics with a certain melting point [4].

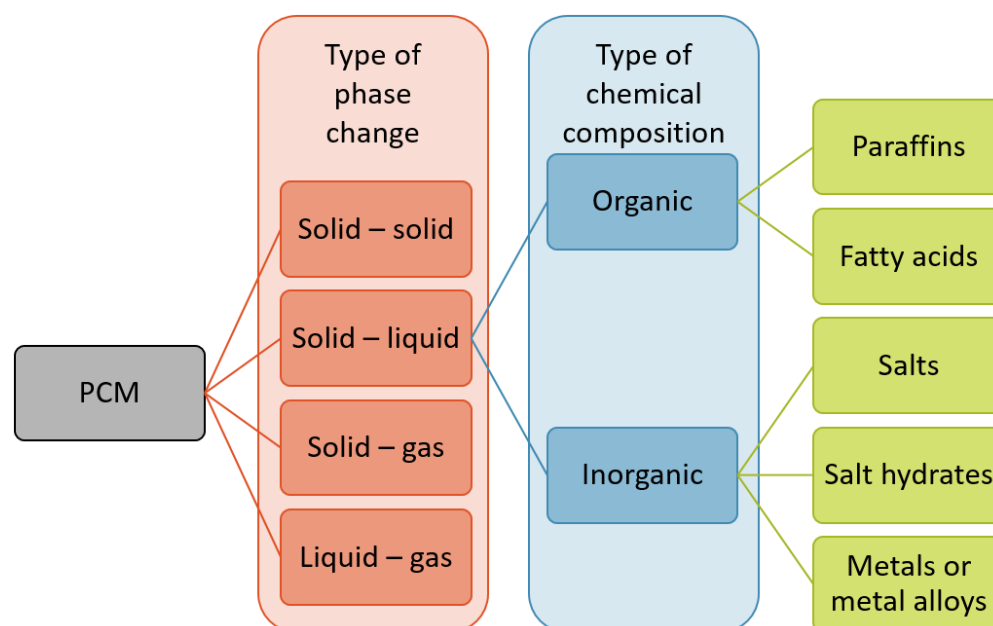


Figure 1. Classification of PCMs. Own elaboration based on [3,4,8,9].

2.2.1. Paraffins

In their pure form, paraffins are saturated hydrocarbons (alkanes) with the molecular formula C_nH_{2n+2} whose melting point increases with increasing number of carbon atoms [5] and can range from about 5.5 °C for hydrocarbon $C_{14}H_{30}$ to about 76 °C for $C_{34}H_{70}$ [6]. In general, paraffins are considered to be safe and non-toxic substances [6,12,13], environmentally benign [4] and without an unpleasant odor [12]. However, those used in practice are relatively cheap paraffins [4,6] with a purity level of technical grade [4,14], which are obtained during petroleum processing [4,12] and are mixtures of hydrocarbons [2,5,12]. Such paraffins may contain environmentally hazardous, toxic and/or carcinogenic substances, such as formaldehyde, vinyl chloride, benzene, toluene, naphthalene or butanone [13,15]. Table 1 summarises the main properties of selected pure paraffins, while Table 2 summarises the main properties of paraffins with technical-grade purity levels. As mentioned at the beginning of this paragraph, pure paraffins are rarely used in practical applications due to their high price, which usually exceeds 25 USD per gram of the material (the price was estimated based on the publicly available data from Merck KGaA). On the other hand, the price of technical-grade paraffins, which are sold as materials for candle making, is usually about 5 USD per kilogram.

Table 1. Overview of selected pure paraffins and their main properties.

Reference Number	Name	Molecular Formula	T_m , °C	L , kJ/kg	ρ , kg/m ³	c_p , kJ/(kgK)
[4,16]	Tetradecane	$C_{14}H_{30}$	5.8–6.0	227–229	759 (l)	n.d. (s); 2.21 (l)
[4,16]	Pentadecane	$C_{15}H_{32}$	9.9–10.0	206	765 (l)	n.d. (s); 2.22 (l)
[4,16]	Hexadecane	$C_{16}H_{34}$	18.0–20.0	216–236	770 (l)	n.d. (s); 2.22 (l)
[4,16]	Heptadecane	$C_{17}H_{36}$	22–22.6	164–214	775 (s)	n.d. (s); 2.23 (l)
[4,16]	Octadecane	$C_{18}H_{38}$	28.0–28.4	200–244	779 (s)	1.91 (s); 2.22 (l)
[4,16]	Nonadecane	$C_{19}H_{40}$	32.0	222	782 (s)	1.92 (s); 2.31 (l)
[4,16]	Eicosane	$C_{20}H_{42}$	36.6	247	785 (s)	1.93 (s); 2.33 (l)
[4,16]	Heneicosane	$C_{21}H_{44}$	40.2	213	788 (s)	1.93 (s); 2.36 (l)
[4,16]	Docosane	$C_{22}H_{46}$	44.0	249	791 (s)	1.93 (s); 2.38 (l)
[4,16]	Tricosane	$C_{23}H_{48}$	47.5	234	793 (s)	1.93 (s); 2.38 (l)
[4,16]	Tetracosane	$C_{24}H_{50}$	50.6	255	796 (s)	1.93 (s); 2.38 (l)
[4,16]	Pentacosane	$C_{25}H_{52}$	53.5	238	798 (s)	1.9 (s); 2.32 (l)

Table 1. Cont.

Reference Number	Name	Molecular Formula	T _m , °C	L, kJ/kg	ρ, kg/m ³	c _p , kJ/(kgK)
[4,16]	Hexacosane	C ₂₆ H ₅₄	56.3	256	800 (s)	1.85 (s); 2.38 (l)
[4,16]	Heptacosane	C ₂₇ H ₅₆	58.8	235	802 (s)	1.92 (s); 2.44 (l)
[4,16]	Octacosane	C ₂₈ H ₅₈	61.2	254	803 (s)	1.91 (s); 2.38 (l)
[4,16]	Nonacosane	C ₂₉ H ₆₀	63.4	239	805 (s)	1.9 (s); 2.45 (l)
[4,16]	Triacontane	C ₃₀ H ₆₂	65.4	252	806 (s)	1.9 (s); 2.46 (l)
[6,16]	Hentriacontane	C ₃₁ H ₆₄	68.0	242	808 (s)	1.89 (s); 2.46 (l)
[6,16]	Dotriacontene	C ₃₂ H ₆₆	69.5	170	809 (s)	1.93 (s); 2.43 (l)
[6,16]	1,2-Tritriacontadien	C ₃₃ H ₆₈	73.9	268	810 (s)	1.88 (s); 2.4 (l)
[6,16]	Tettriacontane	C ₃₄ H ₇₀	75.9	269	811 (s)	1.86 (s); 2.45 (l)

T_m—melting point; L—latent heat of fusion; ρ—density; (l)—liquid; (s)—solid.

Table 2. Overview of selected technical-grade paraffins and their main properties.

Reference Number	Number of C Atoms	T _m , °C	L, kJ/kg	c _p , kJ/(kgK)	ρ, kg/m ³	λ, W/(mK)
[5]	C ₁₅ –C ₁₆	8	153	-	-	-
[3]	C ₁₆ –C ₁₈	22–22	152	-	-	-
[5]	C ₁₃ –C ₂₄	22–24	189	2.1	900 (s); 760 (l)	0.21 (s)
[5]	C ₁₆ –C ₂₈	42–44	189	2.1	910 (s); 765 (l)	0.21 (s)
[5]	C ₂₀ –C ₃₃	48–50	189	2.1	912 (s); 769 (l)	0.21 (s)
[5]	C ₂₂ –C ₄₅	58–60	189	2.1	920 (s); 795 (l)	0.21 (s)
[5]	C ₂₃ –C ₄₅	62–64	189	2.1	915 (s); 790 (l)	0.21 (s)
[5]	C ₂₁ –C ₅₀	66–68	189	2.1	930 (s); 830 (l)	0.21 (s)

T_m—melting point; L—latent heat of fusion; c_p—specific heat capacity; ρ—density; λ—thermal conductivity; (l)—liquid; (s)—solid.

From a practical point of view, an important property of paraffins is their flammability [13,15], which should be taken into account when selecting PCMs for a particular application. An unquestionable advantage of paraffins is that they solidify with little or no supercooling, which means that it is not necessary to use any additives to minimize supercooling [5,7,12,13]. Paraffins in the liquid state are polar substances and are insoluble in water and alcohol [12]. Additionally, paraffins are chemically inert [4,6,12,13] and do not corrode metals [4,7,13,15]. On the other hand, before storing paraffins in plastic containers, it is important to first check how paraffins will affect the plastic in question, as paraffins and plastics are chemically similar, which can lead to the softening of some polymers, especially polyolefins [4,13]. During phase changes of paraffins, a change occurs in their density, and consequently, paraffins in the liquid state occupy a larger volume than paraffins in the solid state. Based on the data in Table 2, it can be seen that the changes in the density of the paraffins presented in the table range from about 10% to 16%. It is also important to note that paraffins are stable [7], meaning that there is no change in their chemical composition [6,12] or phase separation during phase transitions [4,13]. Some authors also report that paraffins can operate for 1000–2000 melting-solidification cycles without significant changes in their properties [4,13]. However, Shukla et al. [17] investigated the properties of three commercially available paraffins. After 600 melting-solidification cycles, no significant change in the melting point was observed, while the heat of phase transition decreased by a maximum of about 20%. According to Sharma et al. [18], the decrease in the heat of phase transition of paraffins might be caused by impurities in the paraffins. Also, paraffins can oxidize when exposed to air, which can result in the formation of undesirable compounds like ketones, aldehydes and carboxylic acids and lead to a loss of heat capacity and a change in phase transition temperature [14]. An additional issue regarding the long-term stability of PCM is the mass loss, which can be observed when PCM is exposed to the environment [19]. Therefore, more research is necessary on the effective encapsulation methods and the long-term stability of PCMs.

One of the main disadvantages of paraffins is their low thermal conductivity coefficient [4,6,13]. As a result, the phase transition may take a long time, which may limit the range of potential applications for paraffin-based PCMs. In order to improve the thermal conductivity, various substances with high thermal conductivity coefficients can be added to paraffins [20], such as carbon fibers [21], metal foams [22], carbon-based nanomaterials [23] or metallic nanomaterials [24].

The abovementioned additives can increase the effective thermal conductivity of PCMs by up to 100 times [25]. Thermal conductivity improvement depends on the type of additive, its shape and size, and mass fraction [26]. It should also be noted that the additives affect other properties of PCMs, including the latent heat of fusion, density and viscosity, the latter two of which are important for convective heat transfer.

Additives usually cause a reduction in the latent heat of fusion, which is associated with a reduction in the mass of the PCM. However, sporadically, intermolecular interactions can occur between the additives and the PCM, which can increase the latent heat of fusion [27].

Additives also affect the density and viscosity of PCMs. These two properties are important during the convective heat transfer that occurs in PCMs when they are in the liquid state. Both the density and viscosity of PCMs increase as the mass fraction of the additive increases [27], which may hinder the convective heat transfer [28]. Additionally, the tendency of the additive particles to form agglomerates and sedimentation also increases with the increasing mass fraction. To avoid the agglomeration and sedimentation of the particles, surfactants like sodium dodecylbenzene sulphonate (SDBS), cetyltrimethylammonium bromide (CTAB), sodium stearyl lactylate (SSL), polyvinyl pyrrolidone (PVP), tween-80, Triton X100, and tetra methylethylenediamine (TEMED) can be added to the PCMs [27,29].

2.2.2. Fatty Acids

The second group of organic PCMs includes fatty acids with the general formula $\text{CH}_3(\text{CH}_2)_{2n}\text{COOH}$ [5,6]. The latent heat of the fusion of fatty acids is comparable with the latent heat of the fusion of paraffins [4–6] and increases with increasing number of carbon atoms in the acid molecule [4,30]. Table 3 summarises the fatty acids that are most commonly considered as PCMs. The prices of those fatty acids vary in the range of about 40–75 USD per kg, considering the purity level of $\geq 98\%$ (estimated based on the publicly available data from Merck KGaA, Darmstadt, Germany).

Table 3. Overview of selected fatty acids and their main properties.

Reference Number	Name	Molecular Formula	T_m , °C	L, kJ/kg	c_p , kJ/(kgK)	ρ , kg/m ³	λ , W/(mK)
[5]	Caprylic acid	$\text{C}_8\text{H}_{16}\text{O}_2$	16.5	149	-	1033 (s), 862 (l)	0.148 (l)
[5,30]	Decanoic acid	$\text{C}_{10}\text{H}_{20}\text{O}_2$	31.5	153	1.9 (s), 2.1 (l)	1004 (s), 886 (l)	0.149 (l)
[5,30]	Lauric acid	$\text{C}_{12}\text{H}_{24}\text{O}_2$	42–44	178	1.7 (s), 2.3 (l)	1007 (s), 870 (l)	0.147 (l)
[5,30]	Myristic acid	$\text{C}_{14}\text{H}_{28}\text{O}_2$	54	187	1.6 (s), 2.7 (l)	990 (s), 861 (l)	0.150 (l)
[5,30]	Palmitic acid	$\text{C}_{16}\text{H}_{32}\text{O}_2$	63	187	1.9 (s), 2.8 (l)	989 (s), 847 (l)	0.165 (l)
[5]	Stearic acid	$\text{C}_{18}\text{H}_{36}\text{O}_2$	70	203	2.35 (l)	941 (s)	0.172 (l)
[4]	Arachidic acid	$\text{C}_{20}\text{H}_{40}\text{O}_2$	74	227	-	-	-

T_m —melting point; L—latent heat of fusion; c_p —specific heat capacity; ρ —density; λ —thermal conductivity; (s)—solid; (l)—liquid.

In general, the properties of fatty acids are similar to those of paraffins [31]. Fatty acids also show no or little supercooling [4–6,14,31,32], and there is no chemical decomposition or phase segregation during phase transition [8,31,32]. Fatty acids are flammable [13,14] and may have an unpleasant odor [4,33] but are non-toxic [14,32]. Changes in the volume of fatty acids during phase transitions are considered to be small [4], about 10% [32]. However,

based on the data shown in Table 3, it can be concluded that fatty acids decrease in density during the transition from the solid to the liquid state and thus increase in volume by about 11–17%.

Fatty acids are considered to be stable after multiple melting-solidification cycles, although Sharma et al. [18] showed that after 1500 melting-solidification cycles, the heat of fusion of stearic acid decreased by about 20%. The presence of impurities in the stearic acid was cited as the reason for this phenomenon. On the other hand, Sar and Kaygusuz [34] investigated the properties of stearic, palmitic, myristic and lauric acids after multiple melting-solidification cycles and the effects of these materials on 304L stainless steel, C20 steel, aluminum and copper. The results showed that the abovementioned fatty acids are stable (their properties do not change) up to 410 melting-solidification cycles. However, once this number is exceeded, the phase transition heat of the acids decreases, and after 910 melting-solidification cycles, the latent heat of stearic, palmitic, myristic and lauric acids decreased by 31.9%, 17.8%, 17.1% and 37.2%, respectively. Regarding the corrosivity of fatty acids, no signs of corrosion were observed on stainless steel or aluminum, while copper was slightly corroded after exposure to each of the fatty acids tested. Also, Ferrer et al. [35] demonstrated that stainless steel and aluminum were resistant to fatty acids while copper gets slightly corroded.

2.2.3. Salt Hydrates

Of the group of inorganic materials, salt hydrates, especially sulfates, phosphates, nitrates and carbonates [36], are most commonly considered as PCMs [31]. The temperature range of phase transitions of salt hydrates is quite wide, from a few degrees to approx. 100 °C [36], and is therefore larger than that of organic materials [37]. Additionally, salt hydrates are characterized by a higher thermal conductivity coefficient compared with organic materials [13,15,31,36,37], as well as a higher heat capacity per unit volume [13,37]. Furthermore, some researchers report that salt hydrates are cheaper than organic materials [13,15,37] and are also non-flammable [13], but after many cycles of melting and solidification, their heat of phase transition may decrease [13,38], and the phase transition temperature may also change [31]. Table 4 presents selected salt hydrates and their main thermophysical properties. It should be noted, however, that their eutectics are often used instead of pure salt hydrates, the composition of which is chosen to obtain a PCM with the desired melting point [38]. Choo and Wei [39] estimated that the price of salt hydrates varies in the range of 1–20 USD per kilogram.

Table 4. Overview of selected salt hydrates and their main properties.

Reference Number	Molecular Formula	T _m , °C	L, kJ/kg	c _p , kJ/(kgK)	ρ, kg/m ³	λ, W/(mK)
[4]	LiCl ₃ ·3H ₂ O	8	253	-	1450 (s)	-
[31]	ZnCl ₂ ·3H ₂ O	10	-	-	-	-
[38]	KF·4H ₂ O	19	231	1.84 (s)	1450 (s)	-
[4]	CaCl ₂ ·6H ₂ O	28–30	190–200	-	1800 (s)	1.088 (s), 0.540 (l)
[4]	LiNO ₃ ·3H ₂ O	30	256	-	-	-
[4]	CaBr ₂ ·6H ₂ O	34	115.5	-	2190 (s)	-
[4]	Zn(NO ₃) ₂ ·6H ₂ O	36	146.9	-	-	0.464 (l)
[38]	Ca(NO ₃) ₂ ·4H ₂ O	42.4	-	1.46 (s)	-	-
[38]	Na ₂ S ₂ O ₃ ·5H ₂ O	49	220	-	1750 (s)	1.46 (s)
[38]	Ni(NO ₃) ₂ ·6H ₂ O	57	169	-	-	-
[38]	MgCl ₂ ·4H ₂ O	58	178	-	-	-
[38]	Ba(OH) ₂ ·8H ₂ O	78	266	-	-	-
[38]	Mg(NO ₃) ₂ ·6H ₂ O	89.9	163	0.669 (s)	1636 (s)	1.81 (s)
[38]	Al ₂ (SO ₃) ₂ ·12H ₂ O	112	-	-	-	-
[4]	MgCl ₂ ·6H ₂ O	117	169	-	-	-

T_m—melting point; L—latent heat of fusion; c_p—specific heat capacity; ρ—density; λ—thermal conductivity; (s)—solid; (l)—liquid.

The general chemical formula of salt hydrates can be written as $AB \cdot nH_2O$, where AB stands for salt and n is the number of water molecules [4,38]. During the phase transition of salt hydrates, their dehydration occurs, and either a hydrate with a lower number of water particles and water or an anhydrous salt and water is formed [2,4,38]. This can result in what is known as incongruent melting, meaning that the amount of water formed during dehydration is not sufficient to dissolve the salt, resulting in differences in material density, phase separation and ultimately salt sedimentation in the reservoir [2,4,13,38]. Phase separation is a common problem with salt hydrates [9,13,31,36,37], but it can be inhibited by adding thickening agents to salt hydrates [4,9,13,38]. These substances include, e.g., cellulose, polyethylene glycol or titanium oxide nanoparticles [38]. Other methods to reduce phase separation include material mixing [13,36], material encapsulation [9,13,36,37] or adding excess water to the salt hydrate—this is the simplest method to reduce phase separation, but it causes a significant increase in the mass and volume of the heat storage [40]. A detailed review of the methods used to reduce phase separation in PCMs was compiled by Zhao et al. [41]. Currently, the addition of thickening agents is considered to be the best method of reducing phase separation due to its simplicity and effectiveness [40].

A second common disadvantage of salt hydrates is their high degree of supercooling [2,4,9,13,31,36–38], which is why nucleating agents must be added to salt hydrates [2,4,9,36,38]. For instance, the hydrate with the formula $Na_2SO_4 \cdot 10H_2O$ can be supplemented with sodium tetraborate (commonly known as borax), which reduces the degree of supercooling of the salt hydrate from 15 °C to approx. 3 °C [38]. Other nucleating agents include but are not limited to K_2CO_3 , $BaCl_2$, $BaCO_3$, CaO , $Mg(OH)_2$, SiO_2 , $SrCO_3$ and many others, which can be found in [42]. The literature also indicates that the addition of nanomaterials [38], encapsulation [2] or the use of containers with rough walls on the side of the PCM [9] can effectively reduce the degree of supercooling. Nonetheless, the addition of nucleating agents is the most effective and the cheapest method since a low amount of the nucleating agent (even below 1 wt.%) can reduce the supercooling degree by more than 90%, maintaining the PCM's latent heat of fusion [42]. Detailed information on the techniques used in supercooling reduction can be found in Zahir et al. [43].

Most salt hydrates that are taken into consideration as potential PCMs can be stored in plastic tanks without the risk of tank damage [2]. On the other hand, salt hydrates are generally corrosive to metals [13,15,37], mainly to copper and aluminum [15,44], while stainless steel can be used with almost all salt hydrates [15,44]. Nevertheless, the problem of the incompatibility of salt hydrates with tank materials can be solved by creating salt hydrate capsules and then placing them in tanks made of any material without risk of corrosion [37].

All of the issues raised in the previous three paragraphs must be taken into account in the context of the long-term stability of salt hydrate PCMs. Generally, the majority of studies on the long-term stability of PCMs are limited to only 200 thermal cycles [45]. Therefore, there is a need for further studies on this issue, which is crucial in determining the actual applicability potential of salt hydrate PCMs.

Different salt hydrates are toxic to humans to varying degrees [13]. For example, $CaCl_2 \cdot 6H_2O$ can cause burns in contact with the skin [13]. $MgCl_2 \cdot 6H_2O$ can cause eye and skin irritation and respiratory problems [13]. Thus, it is advisable to read their technical data sheets carefully before contact with salt hydrates [13].

2.2.4. Salts

The group of salts as inorganic PCMs includes chlorides, fluorides, nitrates, carbonates and sulfates, but instead of pure salts, their eutectics are more commonly used as PCMs [9,36].

Salts have similar properties to salt hydrates; they are also characterized by a high heat of phase transition [36], higher than that of organic materials [37]. Furthermore, their melting point range is very wide [36,37], from about 200 °C to over 1500 °C (see Table 5). For this reason, salts are used to store high-temperature heat [9], e.g., waste heat from industrial processes or from concentrated solar power (CSP) plants [36].

Table 5. Overview of selected salts and their main properties.

Reference Number	Molecular Formula	T _m , °C	L, kJ/kg	ρ, kg/m ³
[4]	AlCl ₃	192	280	-
[4]	LiNO ₃	250	370	-
[9]	NaNO ₃	306	182	2260 (s), 1908 (l)
[9]	KNO ₃	334	266	2109 (s), -
[9]	LiI	458	109	3490 (s), 3109 (l)
[9]	LiBr	550	203	3460 (s), 2528 (l)
[9]	LiCl	610	441	2070 (s), 1502 (l)
[9]	FeCl ₂	677	338	3160 (s), -
[9]	MgCl ₂	714	454	2320 (s), 1680 (l)
[9]	KCl	771	353	1980 (s), 1527 (l)
[9]	NaCl	802	482	2160 (s), 1556 (l)
[9]	Na ₂ SO ₄	884	165	2680 (s), -
[9]	NaF	996	794	2558 (s), 1948 (l)
[9]	K ₂ SO ₄	1069	212	2660 (s), -
[9]	MgSO ₄	1130	122	2660 (s), -
[9]	MgF ₂	1263	938	3150 (s), -
[9]	CaCO ₃	1330	142	2930 (s), -
[9]	CaF ₂	1418	391	3180 (s), 2520 (l)
[9]	BaSO ₄	1512	188	4500 (s), -
[9]	SrSO ₄	1605	196	3960 (s), -
[9]	KNO ₃	334	266	2109 (s), -
[9]	LiI	458	109	3490 (s), 3109 (l)

T_m—melting point; L—latent heat of fusion; ρ—density; (s)—solid; (l)—liquid.

Compared with organic materials, salts are cheaper [37] and have a slightly higher thermal conductivity coefficient [36] but are more corrosive to metals [9,36,46]. Copper, aluminum, and carbon steel containers are not recommended for salt storage [35,47]. However, in most cases, salts do not corrode stainless steel [35,47] and can be stored in plastic tanks, such as polypropylene (PP), polystyrene (PS), poly(ethylene terephthalate) (PET), or high-density polyethylene (HDPE) [47]. Another method to avoid corrosion of metal tanks is to pre-encapsulate the salt and place the capsules in the tanks [37]. Encapsulation can additionally improve the thermal conductivity coefficient of the salt and reduce phase separation [37] and the hygroscopic properties of the salt [48].

Another disadvantage of some salts is that they cause a high degree of supercooling [9,36,37,48,49] and a significant volume change during phase transition [9,37,46], which can exceed 10% [37,49]. Salts are usually non-flammable and are generally considered non-toxic materials, but it is advisable to check their technical data sheets carefully before encountering these materials [13].

2.2.5. Metals or Their Alloys

Metals or metal alloys are not often considered PCMs due to their high mass, comparatively low latent heat of fusion per unit mass and high melting temperatures [9,36,37]. Metals and their alloys have a low specific heat [9,36], but their heat of fusion per unit volume usually significantly exceeds that of organic materials, as well as salts and salt hydrates [9,36]. Other advantages of metals or their alloys include the lack of supercooling [37,46], lack of phase separation [37], small volume changes during phase transformation [8,37,46] and a relatively high thermal conductivity coefficient [9,36,37,46], which

means that there is no need to intensify the heat transfer as was the case with organic materials or salts and their hydrates [9,36]. The range of melting points of metals and their alloys is very wide, which makes them suitable for high-temperature heat storage [36]. Table 6 presents the properties of selected metals and metal alloys that are considered as PCMs.

Metals and their alloys tend to be more expensive than salt hydrates and salts [36,46] but generally do not corrode tanks [37,46]. Nevertheless, metals are electrically conductive [8], and different metals have different chemical activities, as a result of which chemical reactions can occur between them [36].

Table 6. Overview of selected metals or their alloys and their main properties.

Reference Number	Metal or Metal Alloy	Content, wt. %	T _m , °C	L, kJ/kg	c _p , kJ/(kgK)	ρ, kg/m ³	λ, W/(mK)
[37]	Ga	n.d.	29.8	80	0.37	n.d.	29.4
[37]	Rb	n.d.	38.9	26	0.363	n.d.	29.3
[37]	Li	n.d.	186	434	4.39	n.d.	41.3
[9]	Zn/Mg	53.7/46.3	340	185	n.d.	4600	n.d.
[9]	Al/Si/Sb	86.4/9.4/4.2	471	471	n.d.	n.d.	n.d.
[9]	Al/Cu/Mg	60.8/33.2/6.0	506	365	n.d.	n.d.	n.d.
[46]	Al/Si	12/86	576	560	1.038 (s), 1.741 (l)	2700 (s)	160 (s)
[46]	Zn/Cu/Mg	49/45/6	703	176	0.42 (s)	8670 (s)	n.d.
[46]	Cu/Si/Mg	56/27/17	770	420	0.75 (s)	4150 (s)	n.d.
[46]	Cu/Si	80/20	803	197	0.50 (s)	6600	371
[46]	Si/Mg/Ca	49/30/21	865	305	-	2250 (s)	n.d.
[46]	Si/Mg	56/44	946	757	0.79 (s)	1900 (s)	n.d.

T_m—melting point; L—latent heat of fusion; c_p—specific heat capacity; ρ—density; λ—thermal conductivity; (s)—solid; (l)—liquid.

2.2.6. Commercially Available PCMs

Sections 2.2.1–2.2.5 discuss substances that could potentially be used for heat storage as PCMs. However, PCM technology is commercially available, and there are a number of companies worldwide specializing in the manufacture of these materials. Table 7 presents a selection of PCM manufacturers and the materials they offer, but it does not list all available materials; it only lists the product lines offered and their melting point and heat of fusion ranges.

Table 7. Overview of selected commercially available PCMs and their main properties.

Manufacturer	Line of Materials	Number of Materials Available in the Line	Type of Materials	Range of Melting Points, °C	Range of Heat of Fusion, kJ/kg
Rubitherm GmbH (Berlin, Germany) [50]	RT-Line	44	Organic	−9 ÷ +100	120–260
Rubitherm GmbH (Berlin, Germany) [50]	SP-Line	22	Inorganic, salt-based	−50 ÷ +90	150–330
PCM Products Ltd. (Yaxley, Cambridgeshire, UK) [51]	PlusIce S	24	Salt hydrates	+8 ÷ +117	100–220
PCM Products Ltd. (Yaxley, Cambridgeshire, UK) [51]	PlusIce A	52	Organic	+2 ÷ 164	120–305
PCM Products Ltd. (Yaxley, Cambridgeshire, UK) [51]	PlusIce E	16	Inorganic	−65 ÷ 0	175–395
PCM Products Ltd. (Yaxley, Cambridgeshire, UK) [51]	PlusIce H	39	Salts	+104 ÷ +885	70–602
Croda (Snaith, UK) [52]	CrodaTherm™	14	Organic	−22 ÷ +60	157–226
Axiotherm GmbH (Eisenberg, Germany) [53]	ATP	14	Organic	+2 ÷ +78	215–275
Axiotherm GmbH (Eisenberg, Germany) [53]	ATS	11	Inorganic	−33 ÷ +115	145–360

Table 7. Cont.

Manufacturer	Line of Materials	Number of Materials Available in the Line	Type of Materials	Range of Melting Points, °C	Range of Heat of Fusion, kJ/kg
Pluss Advanced Technologies B.V. (Gurugram, India) [54]	savE® HS	14	Salt hydrates	−30 ÷ +88	166–350
Pluss Advanced Technologies B.V. (Gurugram, India) [54]	savE® OM	16	Organic	+4 ÷ +68	157–230
Pluss Advanced Technologies B.V. (Gurugram, India) [54]	savE® FS	5	Organic	+4 ÷ +66	140–172
PureTemp LLC (Bloomington, IN, USA) [55]	PureTemp	24	Organic	−37 ÷ +151	145–301
Phase-change Energy Solutions Inc. (Asheboro, NC, USA) [56]	BioPCM	12	Organic	+4 ÷ +29	210–250
Climator (Skövde, Sweden) [57]	ClimSel™	11	Salt hydrates	−21 ÷ +70	116–288

From Table 7, it can be seen that commercially available materials include inorganic materials, among which salt hydrates predominate, as well as organic materials. The available range of melting points of commercial PCMs is between -65°C and $+885^{\circ}\text{C}$.

The exact composition of commercial PCMs is the manufacturer's secret, but some manufacturers state that their materials consist of, e.g., paraffins or fatty acids, or salt hydrates, as well as other substances added to improve the properties of the materials produced [50,51,54], such as reducing the degree of supercooling [51]. Nevertheless, some characteristics of commercially available PCMs correspond to those of the pure substances that are considered as PCMs.

As reported by some manufacturers, some of the commercially available organic PCMs are described as environmentally friendly [56], biodegradable [52,55,56], and non-toxic [50,51,55,56]. In addition, these materials are chemically inert [50] and non-corrosive [52,56]. Some of them are non-combustible [51,54], but there are also materials with a low flammability rating [52,54]. The degree of supercooling of organic materials is low or non-existent [50,53], while the volume changes during phase transitions are about 10% [53] up to 15% [50], depending on the material. The thermal conductivity coefficient of organic materials ranges from approx. $0.2\text{ W}/(\text{m}\cdot\text{K})$ [50,51,53,56] to $0.6\text{ W}/(\text{m}\cdot\text{K})$ [55]. Additionally, these materials have a long service life [50,56] as they can be subjected to 1000–2000 [54] and even 10,000 [55] melting-solidification cycles without losing their thermophysical properties.

Like organic materials, some of the commercially available inorganic PCMs are non-toxic [53]. A characteristic feature of inorganic materials is their hygroscopy [50]. Manufacturers recommend sealing these materials tightly, especially salt hydrates, in tanks to prevent water absorption or evaporation [50,51]. Unlike organic materials, inorganic materials can corrode metals [50,53] while they have the advantage of non-flammability [53,54] and smaller volume changes during phase transitions, about 3–7% [50]. Moreover, their thermal conductivity coefficient is slightly higher than that of organic materials, at approx. $0.4\text{--}0.7\text{ W}/(\text{m}\cdot\text{K})$ [51,54]. On the other hand, inorganic materials have a slightly higher degree of supercooling than organic materials, although the value of supercooling of inorganic materials usually does not exceed 5°C [50]. Additionally, these materials can be subjected to typically 1000–3000 [53,54] cycles of melting and solidification without risk of deterioration of their thermophysical properties, although some manufacturers state that their materials are stable even for up to 10,000 [50] melting-solidification cycles.

3. Applications of PCMs

The applicability of PCMs is determined by their melting point, and based on the range of melting points of PCMs, they can be divided into high-, medium- and low-temperature materials [4]. For the purpose of this paper, it was assumed that high-temperature PCMs

have melting points higher than 100 °C, medium-temperature PCMs between 15 °C and 100 °C while low-temperature PCMs have melting points lower than 15 °C.

Briefly described below are the most common examples of applications of PCMs found in the literature, while a more detailed description of possible applications of PCMs, as well as applications not described in this paper, can be found in the literature [4,7,31,36,48,58–60].

3.1. High-Temperature Applications

3.1.1. Solar Energy

One of the most promising applications of PCMs is the storage of heat from solar energy, with high-temperature applications including heat storage in so-called solar power plants [3]. In thermal-solar power plants, electricity is generated similarly to the classical Rankine cycle except that the fossil fuel-burning boiler is replaced by a steam generator powered by a heating medium heated in collectors by solar energy [48,61], as schematically shown in Figure 2. When solar radiation is available, part of the heating medium heated in the solar collectors in the solar field is diverted to the steam generator and part to the heat store with PCM, which is melting. When solar radiation is not available, the flow of the heating medium through the solar field is cut off, and the medium absorbs heat from the solidifying PCM and then goes to the steam generator (the direction of the medium flow during the discharge of the heat store is marked in green in Figure 2).

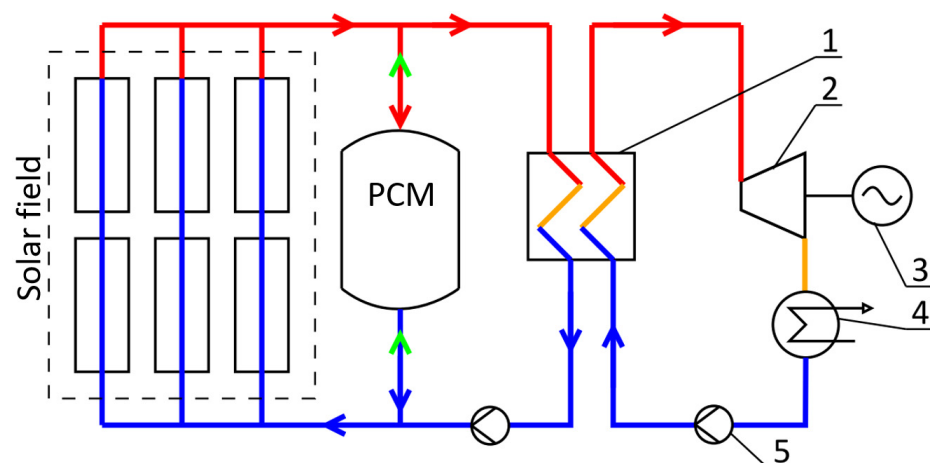


Figure 2. Simplified diagram of a thermal solar power plant with a heat store. 1—steam generator; 2—turbine; 3—generator; 4—condenser; 5—pump. Red line—high temperature; orange line—medium temperature; blue line—low temperature. Own elaboration based on [14,62].

Various technologies are used to focus solar radiation onto the heating medium in order to achieve a sufficiently high heating medium temperature. The most important technologies include parabolic trough collectors (PTCs), solar power towers (SPTs), linear Fresnel reflectors (LFRs) and parabolic dish systems (PDSs) [62,63]. Depending on the solar energy technology used, the temperature range of the heating medium can be 20–400 °C for a PTC, 50–300 °C for an LFR, 300–565 °C for an SPT and 120–1500 °C for a PDS [62,63]. Thus, it can be stated that salt- or metal-based PCMs are most suitable for heat storage in thermal solar power plants due to their melting point.

The advantage of using heat storage with PCMs in thermal solar power plants is that it enables the power plant to operate during periods when solar radiation is unavailable [62] and increases the exergetic and energetic efficiency of the power plant [36]. Additionally, a heat store with PCMs can be almost half the size and 35–43% cheaper than a sensible heat store with the same thermal capacity [64].

3.1.2. Ceramic and Metallurgical Industries

Another example of the use of PCMs with high melting points is heat recovery from the ceramic and metallurgical industries [65]. PCMs in the abovementioned industries can be used to recover and store heat from the waste gases of ceramic and metallurgical furnaces. The accumulated heat can then be further used to pre-heat the combustion air, leading to a reduction in fossil fuel consumption. The range of melting points of PCMs used in the metallurgical or ceramic industry should be around 500–850 °C, so salt- or metal-based materials are suitable [65].

3.2. Medium-Temperature Applications

3.2.1. Construction

Applications of PCMs in the construction industry are usually divided into two groups, i.e., applications in active and passive systems [59,60]. Active solutions involve the use of devices such as a pump or fan, forcing the flow of a heat-transporting medium [59], whereas in passive solutions, heat flow occurs naturally, without the use of additional devices [59].

One example of active systems in which PCMs can be used is the storage of heat or cold in heating or air-conditioning systems for space heating or cooling [66]. For this purpose, containers filled with PCM can be placed in ducts through which water or air flows and the material will undergo appropriate phase transitions depending on the current demand for heat or cooling [66]. An example of such a system, proposed by Zhao and Tan [67], is shown in Figure 3, where a PCM with a melting point of 19–23 °C is placed in the air duct and additionally integrated into the condenser-side refrigeration system. The PCM, in this case, acts as an absorber of heat from the condenser of the air-conditioning system, ensuring a constant condensing temperature of the refrigerant. The PCM, heated and melted during the day, can be cooled at night using cool ambient air, or it can be used for space heating if required.

Another possibility is to incorporate PCMs into the structure of the building, typically in the floors or ceilings, and to route heating or cooling pipes in the PCM [59]. It is also possible to use tanks with PCM instead of traditional water tanks to accumulate heat for heating purposes [60]. Alternatively, the PCM can be incorporated directly with the underfloor heating and cooling systems. In such cases, a layer of PCM is sandwiched between the two layers of mortar and the heating or cooling pipes are placed in the upper layer of mortar [68]. Regardless of the technical solution adopted, i.e., the location of the PCM, the use of PCMs in heating or air-conditioning systems results in a reduction in heating or cooling demand [58] as well as a shift in peak load and a reduction in heating or cooling costs due to the possibility of purchasing cheaper electricity (e.g., at the night rate) to power, e.g., heaters, chillers, pumps or fans [6,58,60].

Other examples of active systems with PCMs are so-called free cooling and the ventilated Trombe wall [59]. In an active free cooling system, a PCM in the form of panels is placed near the ceiling. During the day, the PCM draws heat from the air in the room and is melted. At night, cool atmospheric air is forced into the room by means of a fan, and the PCM solidifies while releasing heat [58]. This solution makes it possible to reduce the temperature fluctuations in rooms [58] and provide thermal comfort to the users [58,59]. A ventilated Trombe wall with PCMs consists of an absorber wall, glazing and an air cavity between the wall and the glazing. The absorber wall is filled with PCMs and has openings at the top and bottom. Solar radiation passes through the glass and heats the air between the wall and the glass, as well as the wall itself, causing the PCM to melt. The air from the room enters through the lower opening in the absorber wall, heats up and flows back into

the room through the upper opening, with air movement being forced by a fan. When solar radiation is not available, the air is heated from solidifying PCMs.

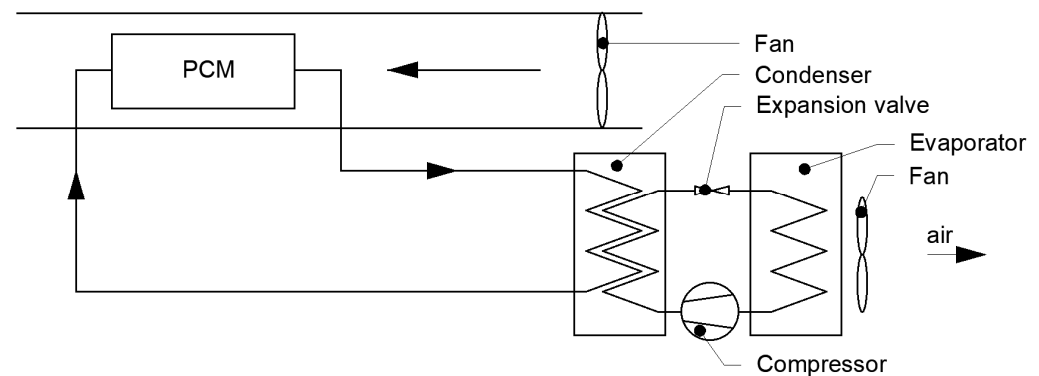


Figure 3. Simplified diagram of an AC system with PCM placed in the air duct. Own elaboration based on [67].

Free cooling and Trombe walls can also occur in passive systems [6,59], i.e., where no device is required to force air movement. In passive systems, PCMs can also be integrated with building components such as walls, floors, ceilings, roofs, or window blinds [6,58–60], which is possible due to the rather high transparency of some PCMs [60]. The integration of PCMs with walls, floors, ceilings or roofs can be carried out in five ways [59], which are described below.

1. Impregnation or direct mixing—a PCM in liquid or powdered form is directly mixed with building materials such as gypsum, concrete or plaster;
2. Absorption of a liquid PCM through the pores of the building material;
3. Use of so-called shape-stabilized PCMs. The manufacture of shape-stabilized PCMs involves the injection of a PCM into a porous material, such as HDPE or expanded graphite. In this case, the melted PCM is retained in the pores of the porous material and does not leak into the environment;
4. Microencapsulation of PCMs—involves surrounding the PCM material with a coating, usually spherical with a diameter of the order of micrometers. The coating does not change the state of aggregation and prevents the PCM from leaking into the environment;
5. Microencapsulation of PCMs—involves embedding the PCM in coatings of a shell material that does not change the state of aggregation in various shapes, e.g., spherical, cylindrical or rectangular. Typically, each of the dimensions of such microcapsules exceeds 1 cm.

The abovementioned encapsulation methods prevent the PCM from leakage, but it must be remembered that there are several challenges related to the encapsulation. Firstly, some shell materials may possess poor chemical, thermal and long-term stability as well as low thermal conductivity [69]. Secondly, each encapsulation technique and shell type has some encapsulation efficiency, which is the ratio of the PCM's mass to the entire mass of the PCM capsule [70]. The encapsulation efficiency is always less than 100%, and since the shell does not change its phase and does not contribute to the latent heat, the latent heat of fusion per the unit mass of the encapsulated PCM is lower as compared to the pure PCM. Another issue with encapsulated PCMs, observed especially in the nano-encapsulated PCMs, is the change in melting point due to the increased pressure and additional decrease in the latent heat of fusion because of the formation of an amorphous layer that does not participate in the latent heat storage [71]. The incorporation of PCMs into building components is intended to raise their thermal capacity and shift the peak heat load, [60],

reduce temperature fluctuations in the rooms [58], and provide thermal comfort to the occupants [58,59,72]. The melting point of PCMs used in the construction industry, and especially in passive systems, should be close to the thermal comfort temperature of the occupants of the rooms [58] and should, therefore, be about 19–24 °C [58], although studies can also be found that report a slightly wider range of 20–30 °C [6].

3.2.2. Solar Energy

One of the most popular applications for PCMs in solar energy generation is the storage of heat from solar collectors for domestic hot water. A PCM can be integrated into a solar thermal collector system in several ways. In the case of vacuum collectors, a PCM can be placed directly in the collector tube (Figure 4a), and the recommended melting point of the material is between 72 °C and 118 °C [58]. The solution for flat plate collectors is similar because it is possible to place a PCM under the absorber (Figure 4b) [66].

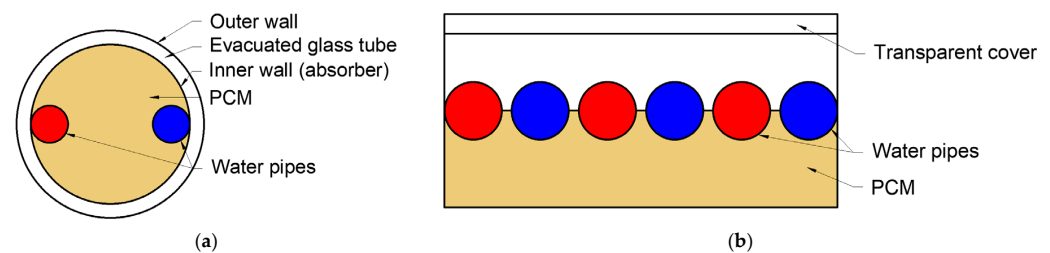


Figure 4. Solar collector with PCM: (a) Vacuum; (b) Flat plate. Own elaboration based on [58,66,73].

Another option is to place capsules with a PCM in a water tank [58], as schematically shown in Figure 5a. Spherical or cylindrical capsules are most commonly used [66], but any shape is acceptable [74]. A fourth way to integrate PCMs into a solar thermal collector system (Figure 5b) is to use two heat storage tanks, a water tank and a tank with a PCM [66,73,74]. In this case, when solar radiation is available, then the heating medium from the solar collectors flows partly into the water tank and partly into the tank with the PCM causing its melting. Then, when solar radiation is not available, the flow of the heating medium through the solar collectors is cut off, and the medium circulates only between the tank with the PCM and the water tank, absorbing heat from the PCM and transferring it to the water. The recommended melting point for PCMs placed in such tanks is between 50 °C and 70 °C [74]. Although the examples described relate to hot water systems integrated with solar thermal collectors, similar solutions can be used when conventional fossil fuel boilers are the heat source instead of solar thermal collectors. The use of PCMs in domestic hot water preparation systems with solar collectors as well as with conventional heat sources makes it possible, first of all, to reduce the volume of heat storage tanks [74] and thus also to reduce the cost of such tanks [66]. In addition, it contributes to the reduction of fossil fuel consumption and greenhouse gas emissions thanks to the possibility of using solar energy [66].

More and more commonly, solar collectors are selected as a source of heat to feed adsorption chillers [75]. Also, in this application, PCMs can be used to store heat from solar collectors during the day and feed the adsorption chiller at night. A simplified schematic of a solar thermal collector system with a heat store in the form of PCMs and an adsorption chiller is shown in Figure 6. During the day, heat from the solar collectors is transferred partly to the heat store with PCMs (store charging) and partly to the water tank. The unloading of the heat store with PCMs takes place at night and involves the transfer of heat from the PCM heat store to the water tank. Subsequently, water from the water tank can be heated in an additional heat source (e.g., electric heaters) if necessary. Finally, the heated water enters the adsorption chiller, which serves as the heat source for the desorption

process. By using a heat store in the form of PCMs, it is possible to extend the operating time of the chiller and reduce the energy consumption of the additional heat source [75]. The required minimum temperature of the heat source feeding the adsorption chillers is approx. $65\text{ }^{\circ}\text{C}$, and therefore, the melting point of the PCMs performing the function of a heat store for the adsorption chillers should be greater than $65\text{ }^{\circ}\text{C}$ [75]. Although the heat source feeding the adsorption chiller in the installation shown in Figure 6 is a solar collector, an alternative heat source for such chillers can also be, e.g., waste heat from industrial processes, which can also be stored in PCMs.

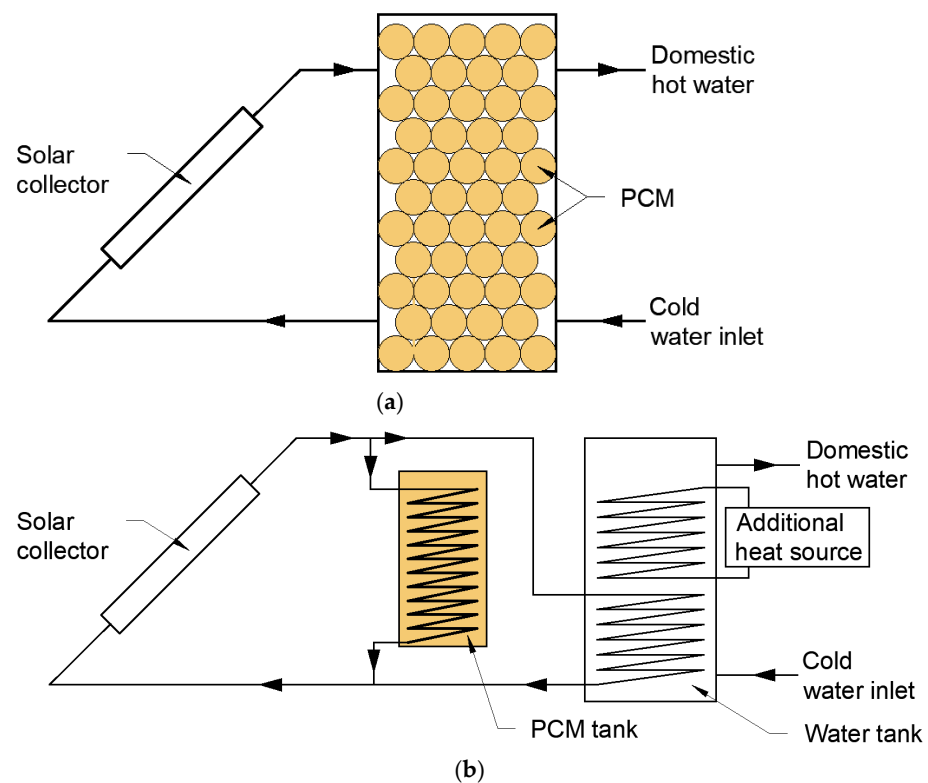


Figure 5. Examples of solar collector systems with PCMs are (a) water tanks with PCM capsules and (b) separate tanks for PCM and water. Own elaboration based on [66,74].

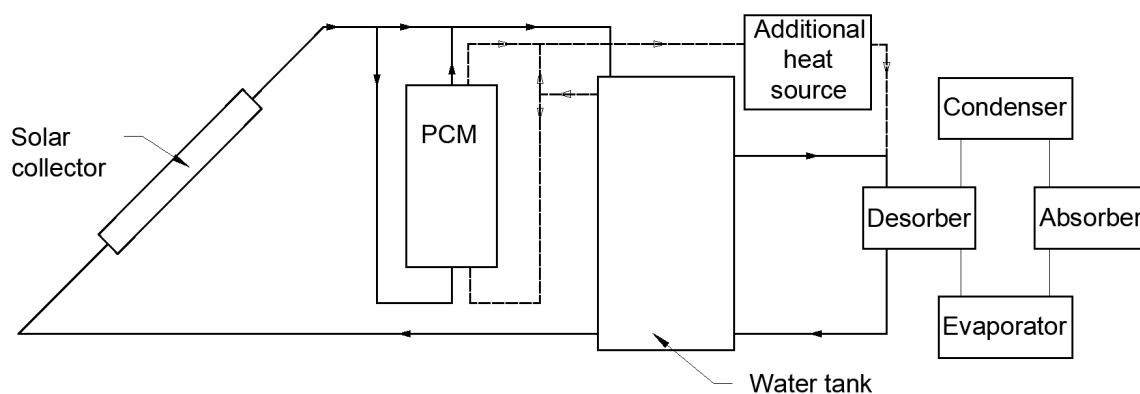


Figure 6. Solar thermal collector system integrated with a heat store and an adsorption chiller. Solid line—heat storage charging; dashed line—heat storage discharging. Own elaboration based on [75].

PCMs can also be used in air solar collectors. In this case, PCMs can be placed directly in the collectors or in external tanks [76], and the recommended range of melting points of PCMs used for air collector heat storage is between $36\text{ }^{\circ}\text{C}$ and $78\text{ }^{\circ}\text{C}$ [77]. Airflow through air collectors can be natural or forced using a fan [73], and such collectors can be used for

heating spaces, dehumidifying air as well as drying fruit or vegetables [73,76,77], and they can also be integrated into greenhouses [76]. However, PCMs can also be placed directly in greenhouses to store heat during the day and release it at night [76], providing a constant and higher temperature [78]. The most common PCM used in greenhouses is $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ with a melting point of 28–30 °C.

Other examples of the use of PCMs for solar energy storage include solar water distillers and solar cookers. The main structural components of a solar water distiller are a tank and usually a glass cover inclined at an angle, usually equal to the latitude of the locality in which the distiller will operate. The bottom of the tank can be, e.g., a black-painted metal sheet performing the function of an absorber. The tank is filled with a thin layer of water, and incident solar radiation causes the water to heat up and evaporate, which then condenses on the lid and drains into the tank. A PCM in solar water distillers can be placed on the absorber in cylindrical [79] or spherical tanks [80] or as a layer under the absorber plate [81]. The melting point of the PCMs used in solar water distillers should be close to the maximum temperature of the water in the distiller [81,82] and should usually be between 40 °C and 60 °C [83]. PCMs can improve the efficiency, i.e., the daily amount of distillate obtained by up to 100% [84]. In solar cookers, PCMs are used to extend the operation time [73]. Data in the literature give different ranges of melting points for PCMs used in solar cookers, and some sources report a range of 55–118 °C [73] and others even up to 253 °C [85].

3.2.3. Mobile Thermal Energy Storage

Mobile thermal energy storage (MTES) is another possible application of PCMs. MTES can be a promising alternative to traditional heat supply technologies [86]. In traditional heat supply technologies, heat is generated in heating or heat and power plants, and then it is transferred by heat transfer fluid (usually water) through a dedicated district heating network (a system of insulated pipes) to the end users, e.g., households. However, the construction of district heating networks is not always technically and/or economically feasible. In such cases, the MTES technology can be an alternative. The idea of MTES is to charge a container filled with PCM, transport the MTES by road and discharge it at the heat consumer, which is schematically shown in Figure 7.

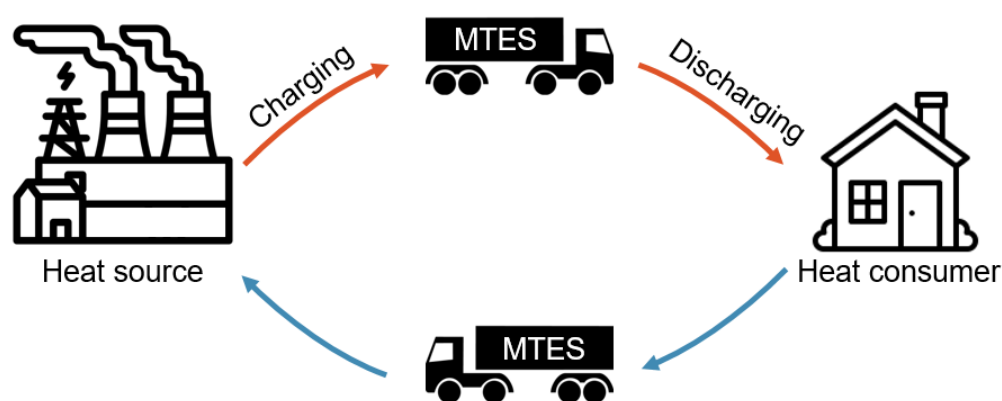


Figure 7. Operating principle of mobile thermal energy storage (MTES).

As the MTESs are usually used to supply heat for space heating and/or domestic hot water preparation, the erythritol (melting temperature of 118 °C) and sodium acetate trihydrate (melting temperature of 58 °C) are usually used in the MTES [87]. The MTES technology allows for, inter alia, recovering waste heat, transporting it, and using it in remote and isolated areas [88], leading to improved energy efficiency and carbon neutrality [89].

3.2.4. Automotive Industry

PCMs can have many applications in the automotive industry. In the case of cars with internal combustion engines, PCMs can be used to recover and store heat from the exhaust gas, which can then be used to, e.g., pre-heat oil in the gearbox or to heat the cabin [90]. Heat recovery and storage can be particularly important for cars with compression-ignition engines that are operated in countries where temperatures fall below 0 °C [91]. The occurrence of such low temperatures makes it necessary to pre-heat engines, which leads to increased fuel consumption [91]. In a study conducted by Kauranen et al. [91], the use of a PCM with a melting point of 75 °C to recover and store heat from the exhaust gas made it possible to avoid additional fuel combustion to pre-heat the engine.

On the other hand, hybrid or electric vehicles are becoming increasingly popular. For such vehicles, it is necessary to keep the temperature of the batteries within a well-defined range, which can be achieved, e.g., by using PCMs with a melting point of about 40 °C [92]. The use of PCMs to maintain a constant battery temperature in electric vehicles can extend battery life by up to 100% and can also increase the driving range of electric vehicles [92].

3.2.5. Road Infrastructure

PCMs can also have applications in road infrastructure. They can be added to asphalt during road construction [93] and to concrete or asphalt pavements [94]. This allows the asphalt or concrete to maintain a constant temperature and prevents it from cracking and forming ruts [93]. It is essential, however, that the PCM does not leak out of the asphalt or concrete, and therefore, it is not directly mixed into the asphalt or concrete, but shape-stabilized PCMs or capsules are first prepared and then added to the asphalt or concrete [93]. However, the addition of PCMs to asphalt and concrete results in a reduction in their mechanical strength, so the amount of PCM added must be selected accordingly [93,94]. PCMs for road infrastructure applications should have melting points in the range of approx. 1–60 °C [94].

3.2.6. Cooling of Electronic Equipment, Batteries and Photovoltaic Panels

Due to the miniaturization and increasingly compact design of electronic equipment, there is a growing problem of overheating, which can lead to performance degradation and failure [95], as well as discomfort for users [95,96]. In order to protect electronic devices from overheating, PCMs with melting points in the range between 35 °C and 60 °C should be used [96]. Additionally, it is possible to create composite PCMs with a stable shape that can be as thin as 0.4 mm [95] so that electronic devices can be kept compact.

A temperature rise for rechargeable batteries used, e.g., in portable electronic equipment, can result in a reduction in battery capacity, and it is assumed that the temperature of batteries should not exceed 40 °C [96]. To ensure that this temperature is not exceeded, a layer of a PCM can be laid around the battery to absorb the heat from the battery [97]. Bearing in mind that cooling the electronics should proceed at an appropriate rate and PCMs possess low thermal conductivity, heat sinks composed of PCM embedded into a metal foam are gaining more attention [98]. In such cases, the metal foam acts as a thermal conductivity enhancer, intensifying heat transfer.

Increased temperatures also have a detrimental effect on PV panels as their power and voltage decrease as the temperature rises [96]. In this case, a layer of PCM with a recommended melting point of between 20 °C and 40 °C can be laid under PV panels [99].

3.2.7. Textile and Clothing Industry

The application of PCMs in the textile and clothing industry involves incorporating PCMs into the structure of clothing or accessories, such as sleeping bags, blankets, mat-

trusses, pillows or bandages, to maintain the wearer's thermal comfort despite changing weather conditions or changes in physical activity [100], as well as to maintain a specific body temperature of patients during medical procedures [4]. PCMs should maintain a temperature that is comfortable for the wearer, and therefore, their melting point should be in the range between 18 °C and 35 °C [100]. The manufacture of garments and textiles with PCMs can be done in four ways, as described in detail in the paper [100].

3.3. Low-Temperature Applications

3.3.1. Food Storage and Transport

PCMs can be utilized in domestic and industrial refrigerators and freezers used for food storage. PCMs can be integrated into fridge freezers in three ways. Firstly, PCMs with a melting point of approx. 30 °C can be placed on the condenser in the form of, e.g., panels or shape-stabilized PCMs [101], which allows heat to be transferred to the environment at a constant temperature and results in an increase in the COP, a reduction in electricity consumption and a reduction in compressor run time [58,101].

A second possibility is to place PCMs on the evaporator side, which allows for an increase in the COP [58,101,102], a reduction in electricity consumption and a reduction in compressor run time [58,101,103], thus minimizing the risk of compressor failure [103]. Additionally, the phase transition of PCMs allows the refrigerant in the evaporator to absorb more heat at a constant, usually higher, temperature, which raises the COP [58,102,103]. Furthermore, heat transfer is faster between the evaporator and the PCM than between the evaporator and the air [103]. An example of an evaporator or condenser with shape-stabilized PCMs proposed by Cheng et al. [101] is shown in Figure 8.

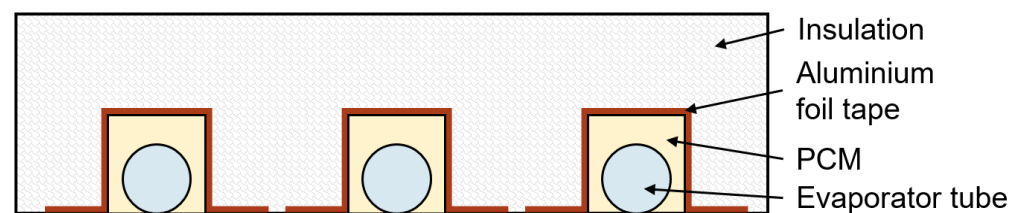


Figure 8. A condenser or evaporator with a PCM can be used in domestic refrigerators. Own elaboration based on [101].

A third technical solution used in refrigerators is the incorporation of PCMs in the enclosures of these appliances [58]. This solution minimizes temperature fluctuations that occur in refrigerator compartments, e.g., due to the opening of the doors [7,58,102,103]. Additionally, compressor run time is reduced as well as electricity consumption [58] while in the event of a power outage, PCMs make it possible to maintain a low temperature in the refrigerator and freezer compartments for a longer time [102,103], thus protecting food products from spoilage. When PCMs are placed in enclosures or on evaporators of refrigerators, the melting point of the PCMs should be equal to the temperature at which the food products in question are required to be stored [103]. The use of PCMs in refrigerators can offer a number of benefits, but a limitation that prevents the large-scale use of refrigerators with PCMs is their high price [58].

PCMs can also be used when transporting food products, both warm and chilled, such as milk or frozen foods [3,7,58]. In this case, as in the case of refrigerators, a PCM can be integrated into the refrigeration system or placed as panels in the walls of the vehicles [58]. Integrating a PCM into the refrigeration system involves connecting the tank (heat exchanger) with the PCM in parallel to the evaporator of the refrigeration system. During periods of low cooling demand, the store with a PCM is charged (material solidification occurs), while during periods of peak load, the cold store is discharged,

resulting in a reduction in the energy consumption of the refrigeration system as well as a reduction in its power and volume [58]. The second option, i.e., placing panels with PCMs in the walls of transport vehicles, allows the temperature inside the vehicle compartment to remain constant for longer periods [58].

A similar result, i.e., maintaining a constant temperature for an extended period, is achieved when a PCM is placed in packaging or containers designed for food storage [58,104]. PCMs increase the heat capacity of such containers and provide better thermal protection for food products [58]. One example of a container containing a PCM can be a vacuum flask in which the space between the walls can be filled with a PCM [7].

3.3.2. Air Conditioning and Refrigeration Systems

PCMs with melting points below 15 °C in air-conditioning and refrigeration systems are usually placed in containers connected in parallel to the evaporator of refrigeration systems, as shown in Figure 9 [31]. When there is little or no demand for refrigeration, the working fluid from the evaporator is directed to the store with a PCM and/or to the fan coil unit, while during times of high demand for cooling, the working fluid absorbs the cold stored in the PCM material and transfers it to the fan coil unit. This technical solution can be used both in conventional refrigeration systems and in, e.g., solar-powered absorption and adsorption chillers [31].

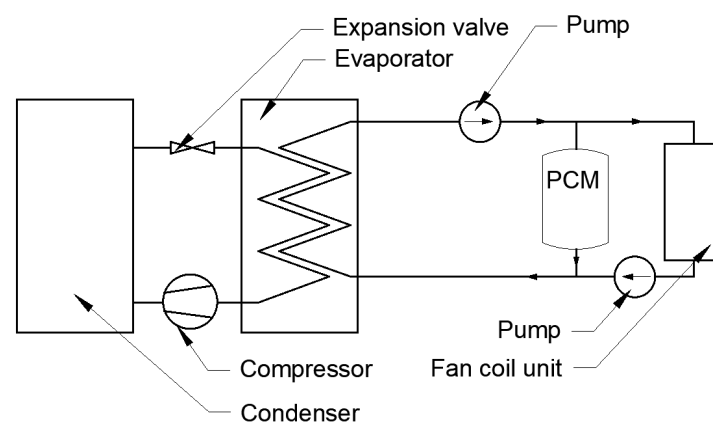


Figure 9. Simplified diagram of a refrigeration system with a cold store in the form of a PCM. Own elaboration based on [31].

However, the parallel connection of the storage tank with a PCM to the evaporator of the refrigeration system makes the whole system more technically complex. Therefore, Mselle et al. [105] designed a heat exchanger, as shown in Figure 10a, that acts as an evaporator and heat store at the same time. A diagram of the connection of this exchanger to the refrigeration system is shown in Figure 10b. After passing through the heat exchanger, the heat transfer fluid is cooled and can be used for cooling, e.g., rooms. By combining the functions of an evaporator and heat store in the heat exchanger design, the entire refrigeration system is simpler and occupies less volume compared with a system with a heat store connected in parallel to the evaporator [105]. Additionally, the PCM ensures a constant evaporation temperature of the working fluid, which leads to energy savings [105]. Such a solution, i.e., a heat exchanger acting as an evaporator and heat store, can be used in air source heat pumps used in temperate climates for space heating [106]. Then, a PCM with a melting point of approx. 6 °C can provide the heat source for the heat pump evaporator [106].

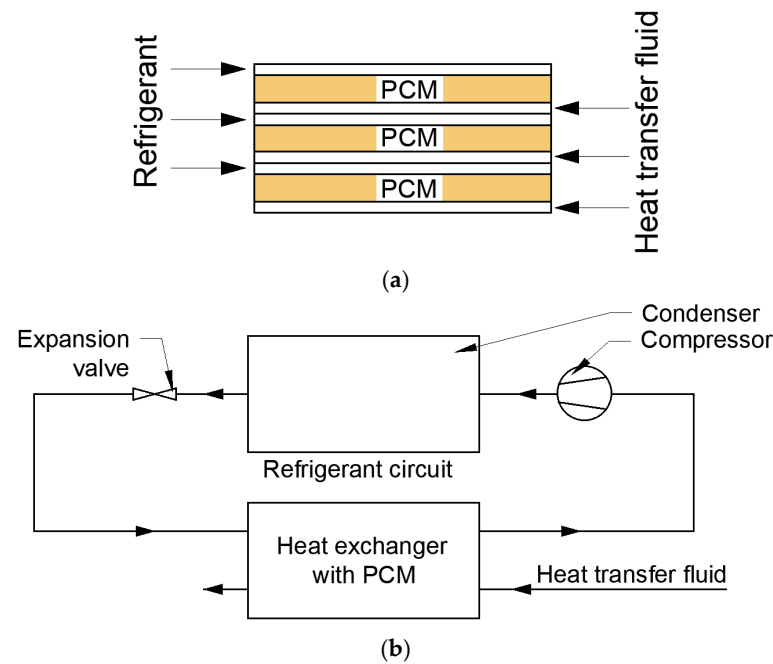


Figure 10. Schematic diagram of: (a) Heat exchanger acting as an evaporator and heat store at the same time; (b) Diagram of refrigeration system with a heat exchanger. Own elaboration based on [105].

Another possible application of PCMs in air-conditioning systems is the creation of so-called PCM slurries [102]. Such a suspension usually consists of a liquid, e.g., water, to which microcapsules with a PCM are added [102]. The advantage of such a slurry is its high heat capacity and the fact that it can act as a heat store and be a heat transfer fluid [102]. It is also worth noting that the phase transition of water into ice [106] can also be used to store cold from air-conditioning or refrigeration systems, and the so-called ice slurry, i.e., a suspension of ice crystals in water or in an aqueous solution of, e.g., sodium chloride, ethanol or ethylene glycol, can be used as a cooling medium [107].

Water-based ice slurry can also be used as a cold store and carrier in air cooling systems in mines instead of the traditionally used water. In large mines, air cooling in the mining areas is provided by coolers to which chilled water is supplied from the surface. Over time, the mining areas are moved significantly further away from the mine shafts, and consequently, the pipelines transporting chilled water to the air coolers take longer. Despite thermal insulation, the lengthening of the pipelines transporting chilled water results in increased thermal losses, which in turn results in an increase in the temperature of the chilled water supplied to the air coolers in the mining areas. Unfortunately, the increase in the temperature of the chilled water at the inlet to the air coolers results in a decrease in the performance of the coolers and an increase in the air temperature in the miners' working area. Currently, temperature hazard is one of the main problems for mines. An example of a solution to this problem is the use of ice slurry instead of chilled water in the central air cooling system of the Bogdanka Coal Mine [108]. Prior to the expansion, the central air-conditioning system of the mine used chilled water with a temperature of approx. 1.5 °C transferred through a network of pipelines to air coolers in the mining areas located at a depth of about 1 km underground. The water was chilled at the surface at a rate of approx. 270 m³/h and flowed through insulated shaft pipelines and further through the Pressure Exchanger System (PES) toward air coolers located several kilometers from the shaft. The return pipeline transported water at a temperature of approx. 20 °C back through the PES and a return shaft pipeline to a water cooling hall located on the surface (closed-circuit cooling). In this hall, the return water was again cooled to a temperature of

approx. 1.5 °C by means of cooling equipment and again transferred through the PES to the underground air coolers. The retrofitting solution applied, in which ice slurry is used instead of chilled water, makes it possible to significantly increase the transport capacity of the ‘cold’ in the cooling system without expanding the existing infrastructure, i.e., without increasing the diameter of the transport pipelines. After the 2021 upgrade, a reduction in the temperature of the inlet fluid from approx. 10 °C (chilled water) to approx. 0 °C (ice slurry) was achieved in the outermost air coolers [88]. After the retrofitting, the system utilized the previously existing cooling infrastructure together with the existing machine hall on the surface, to which a new section responsible for the production of ice slurry was added (Figure 11). The main part of the extension of the cooling system is the system for the production of ice particles, which is located entirely on the surface. The main equipment of the ice particle production system is a vacuum ice particle generator with a concentrator, an ammonia cooling system, an ice slurry mixer, a pumping system and a water filtration system.

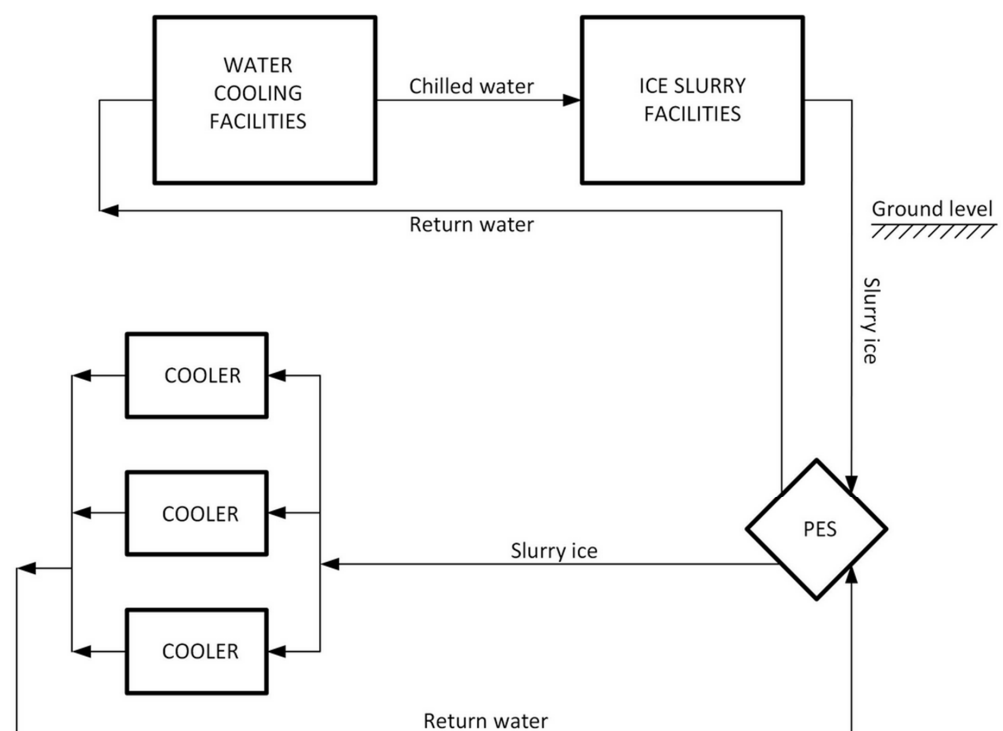


Figure 11. Technological scheme of the slurry ice system for the mine.

The melting point of the PCM used in refrigeration and air-conditioning systems depends on the intended use of the system, but it can be assumed that air-conditioning systems used to provide thermal comfort to people should use PCMs with a melting point of approx. 5–15 °C [31].

The aim of using PCMs in air-conditioning and refrigeration systems is to reduce energy consumption [4,58] and thus also to reduce emissions of carbon dioxide, sulfur dioxide, nitrogen oxides and chlorofluorocarbons [58]. The integration of PCMs into refrigeration and air-conditioning systems also makes it possible to reduce the power and size of refrigeration and air-conditioning systems [4,7,31] and to shift electricity consumption to a cheaper rate (e.g., electricity consumption is reduced during the day and increased at night) [4,6,7,31,58] leading to economic savings [7,31,58]. Additionally, PCMs increase the flexibility of refrigeration systems [7] and can be used as a backup source of cooling, e.g., in the event of a power outage [31]. Compared with cold stores that only use sensible

heat of the storing substance, stores with PCMs have a smaller volume while maintaining a constant thermal capacity [7,58], but their cost is higher [58].

3.4. PCMs in Hybrid Energy Storage Systems

Commitment to sustainable development and climate neutrality is a driving force to the growth of hybrid energy storage systems, i.e., systems that consist of at least two different energy storage technologies [109]. Currently, PCMs are used in hybrid systems with batteries. Such a solution allows for the simultaneous storage of heat and electricity [110]. Also, PCMs are increasingly used in domestic hot water tanks in which both water and PCM store thermal energy, but in two different ways—water in the sensible form, while PCM in the latent form [111,112].

4. Conclusions

This review highlights the significant potential of PCMs as a cornerstone technology in advancing sustainable energy storage solutions. PCMs, leveraging their high latent heat, are uniquely positioned to address energy efficiency challenges in a range of sectors. For example, in solar energy systems, PCMs enable thermal energy storage in both medium- and high-temperature ranges. Applications in concentrated solar power (CSP) plants using molten salts can store energy at temperatures up to 1500 °C, facilitating power generation during non-solar hours and improving operational efficiency.

In building technologies, PCMs are integrated into walls, floors, and ceilings to stabilize indoor temperatures and reduce energy demand for heating and cooling. Advanced systems using encapsulated PCMs in HVAC units further improve thermal comfort and energy efficiency. PCMs with a range of melting points between 19 °C and 24 °C are particularly effective in maintaining comfortable living conditions in passive and active building designs.

The automotive sector benefits from PCM integration in battery thermal management systems, particularly for electric vehicles (EVs). By maintaining battery temperatures within an optimal range (20–40 °C), PCMs enhance performance, extend battery life, and improve safety. Furthermore, PCMs are increasingly used to recover waste heat from internal combustion engines, reducing fuel consumption and emissions in cold climates.

In cold chain logistics and refrigeration, PCMs with melting points between −20 °C and 5 °C ensure temperature stability during the storage and transportation of perishable goods and pharmaceuticals. For instance, PCM-based packaging solutions maintain vaccine stability in regions with limited access to refrigeration, aligning with global health goals.

Emerging applications include electronics cooling, where PCMs prevent overheating of processors and electronic components, and industrial waste heat recovery, where PCMs store thermal energy from exhaust systems and transfer it for pre-heating processes. PCMs are also found in agriculture, e.g., in greenhouses, where they maintain stable temperatures for crop growth by absorbing excess heat during the day and releasing it at night.

Despite these possibilities, challenges such as low thermal conductivity, phase separation, and supercooling remain barriers to broader adoption. Solutions such as encapsulation, hybrid materials, and nano-enhanced PCMs have shown potential to overcome these issues. Additionally, encapsulation methods prevent leakage and ensure durability over thousands of thermal cycles. Also, policy and industry regulations play a crucial role in supporting the widespread adoption of PCMs. Governments can provide grants to organizations and institutions for PCM research and development. Furthermore, some subsidies or tax incentives can be supplied to PCMs-based technologies to be more economically competitive with other thermal energy storage technologies. There is also a need to establish clear guidance for PCM testing and introduce standards of quality assurance.

In conclusion, PCMs are a transformative technology with diverse applications across energy, transport and industry. By enhancing energy efficiency, reducing greenhouse gas emissions, and lowering reliance on fossil fuels, PCMs contribute significantly to global sustainability goals. Future efforts should focus on advancing PCM research, scaling production processes, and integrating PCMs into emerging technologies, ensuring their widespread adoption in the pursuit of a cleaner, more sustainable energy future.

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Abbreviations

The following abbreviations are used in this manuscript:

CSP	Concentrated solar power
EV	Electric vehicle
HDPE	High-density polyethylene
IEA	International Energy Agency
LFR	Linear Fresnel reflector
MTES	Mobile thermal energy storage
PCM	Phase change materials
PDS	Parabolic dish system
PES	Pressure Exchanger System
PET	Poly(ethylene terephthalate)
PP	Polypropylene
PS	Polystyrene
PTC	Parabolic trough collector
RES	Renewable energy sources
SPT	Solar power tower
TES	Thermal energy storage

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